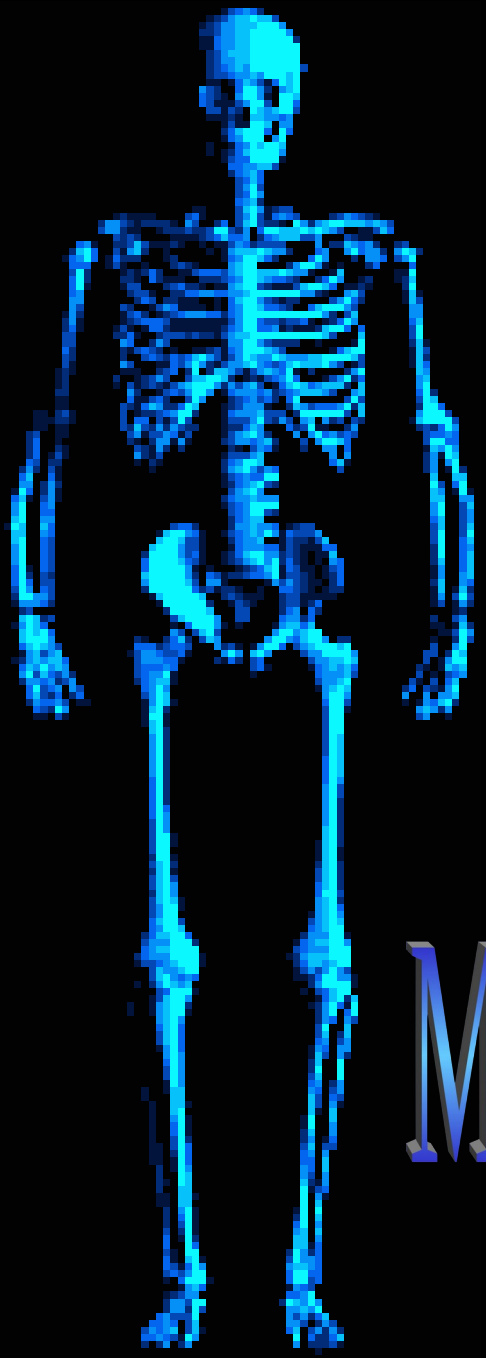
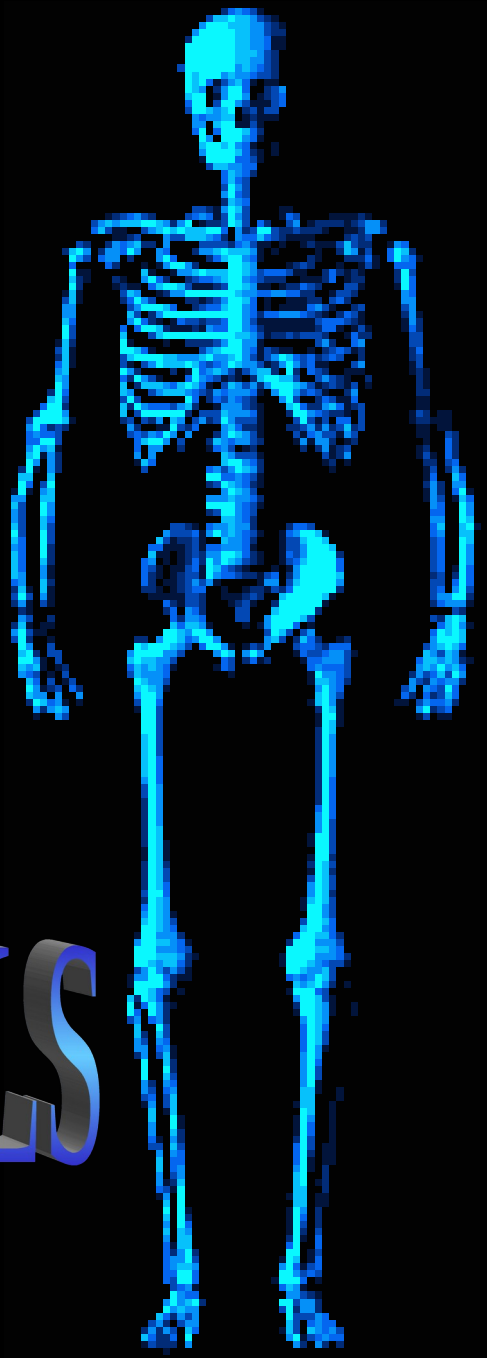




MACROMINERALS

There Are 7 Major Categories of Nutrients:

Introducing Minerals



Minerals



Proteins



Fats



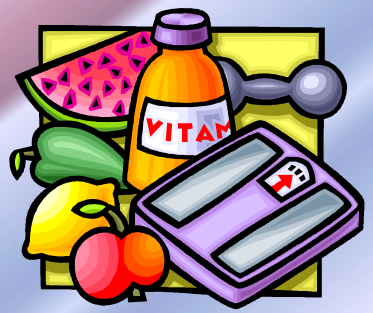
Carbohydrates



Fiber



Water



Vitamins

Minerals are elements that originate in the Earth and cannot be made by living organisms. They are, therefore, inorganic. Plants obtain minerals from the soil, and most of the minerals in our diets come directly from plants, however this varies with geographic locale because of soil mineral content. Some minerals come indirectly from animal sources.…not from the beef of the cow, but from the milk of the cow. Minerals may also be present in the water we drink, but this also varies with geographic locale.

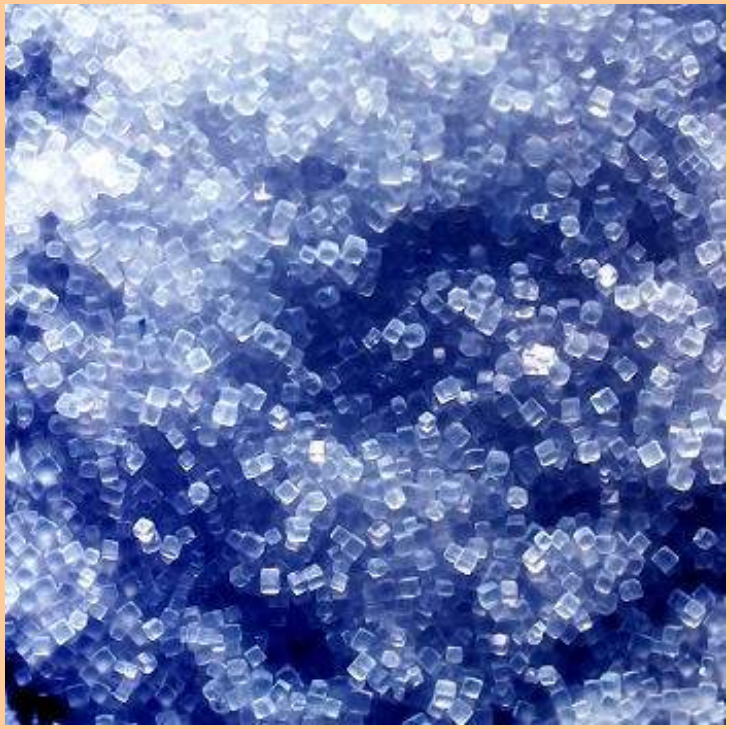
Like vitamins, minerals often serve as coenzymes. They help protein form enzymes that speed up body reactions during digestion and metabolism.

Minerals are sturdy nutrients, meaning they will not be destroyed during most food preparation methods, although some leach out into the water used in various cooking methods.

About 17 minerals are considered 'essential' to humans… required for proper functioning and good health. Those seven needed in amounts of 0.1 gram or more daily are considered major minerals, or MACROMINERALS.

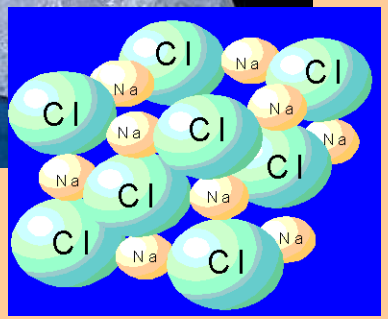
Introducing 'sodium' and 'chloride'...

An ion is an atom or group of atoms that carries a positive or negative electric charge as a result of having lost or gained one or more electrons. Sodium is a ‘cation’ (kăt'ī‘on) or a positively charged ion… NA+. Chloride is an ‘anion’ (ăn'ī'ən) or the negatively charged ion of chlorine… Cl^-. Together they form SALT.









Sodium and chloride are essential macrominerals.

Sources of sodium and chloride...

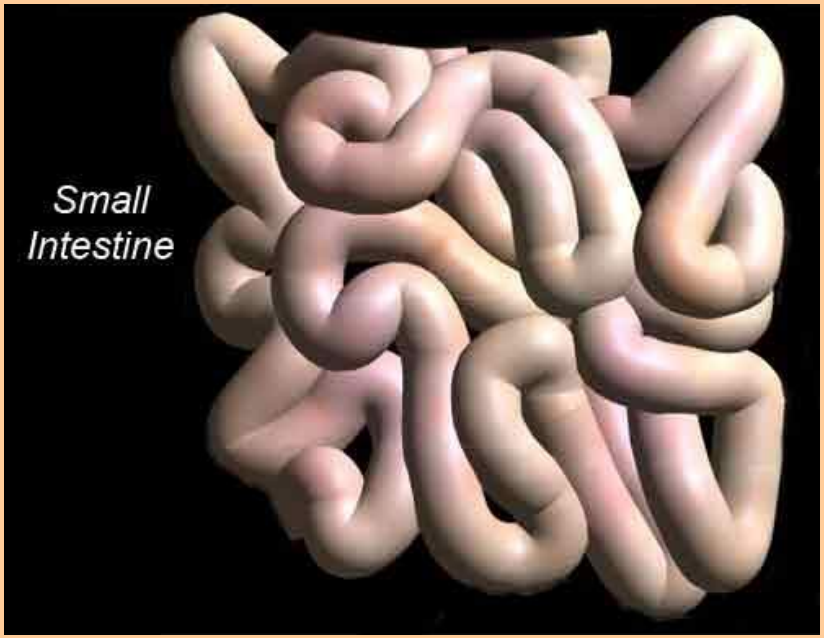
Approximately 75% of your daily salt intake in the U.S. comes from processed and manufactured foods. The remaining 25% comes naturally in foods and from adding salt to foods at the table.

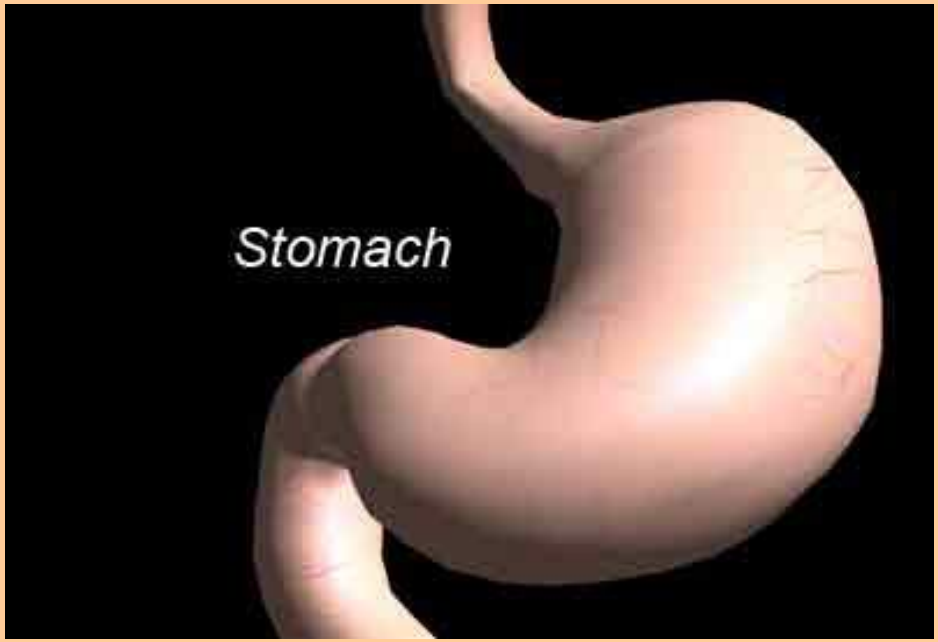


The lowest salt intakes are associated with diets that emphasize unprocessed foods, especially fruits, vegetables, and legumes.

Functions of sodium and chloride...

Absorption of sodium in the small intestine plays an important role in the absorption of chloride, proteins (amino acids), carbohydrates (glucose), and water. Chloride, in the form of hydrochloric acid (HCl), is also an important component of gastric juice, which aids the digestion and absorption of many nutrients. Sodium also helps the body to retain water and prevent dehydration, and may have some activity as an antibacterial.

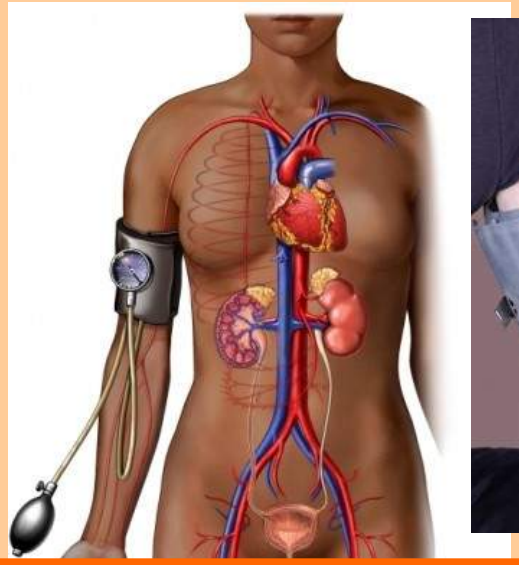




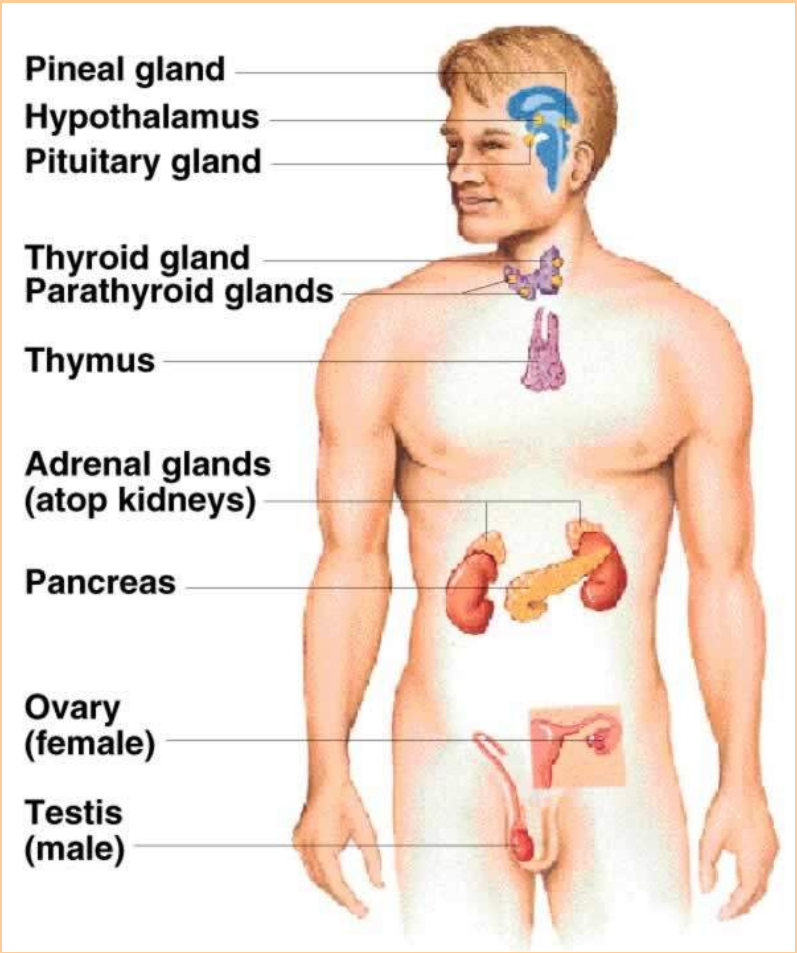
Endocrine glands, such as the pituitary, thyroid, adrenals, and pancreas work with the kidneys to regulate sodium levels. They 'sense' changes in blood pressure and volume. Blood pressure is a measure of the force blood exerts on the walls of blood vessels.

<u>When blood pressure is low</u>, the kidneys hold onto sodium and the body responds by holding onto water. This increases blood volume and blood pressure.

<u>When blood pressure is high</u>, the kidneys release sodium, and the result is water loss in the body. Blood volume and blood pressure decrease.

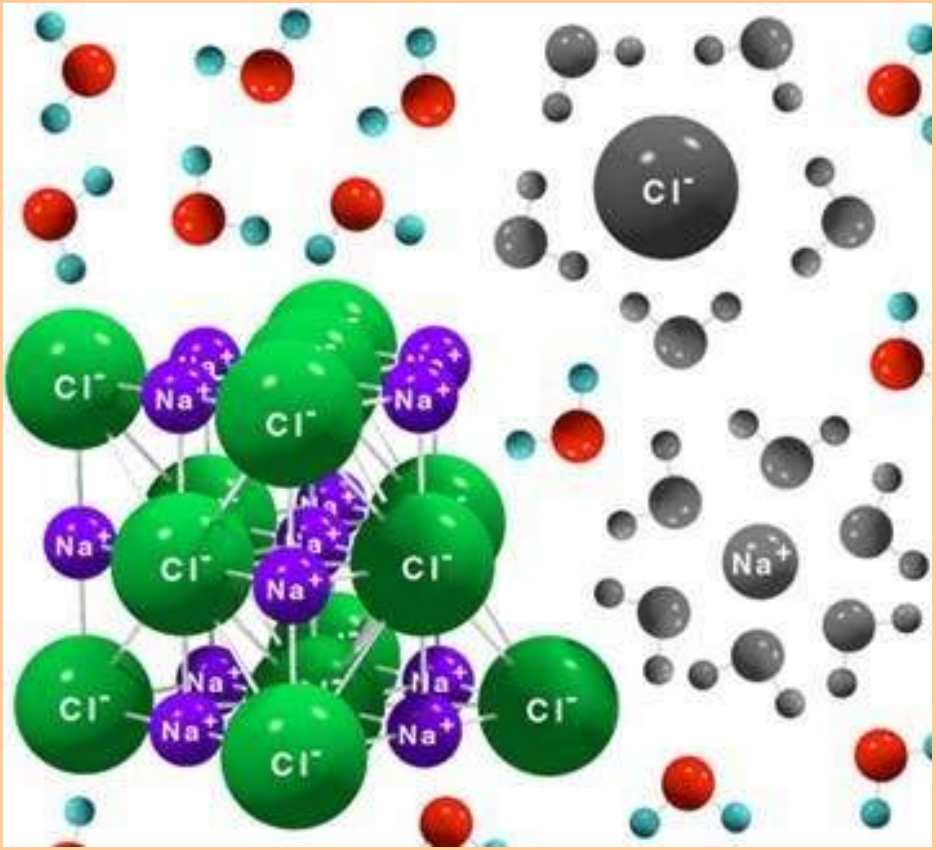






Sodium and chloride are electrolytes...

Sodium and chloride are electrolytes, meaning they are the positively and negatively ionized salts in body fluids, primarily those outside the cells including blood plasma. When sodium is dissolved in bodily fluids, its mild electrical charge is necessary for normal functioning of nerve transmissions and muscle contractions.



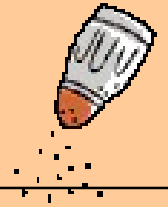
Sodium works hand-in-hand with another mineral, potassium, to maintain the concentration and pH (alkalinity and acidity) balance of intra- and extra-cellular fluids. This balance is referred to as 'membrane potential', and will be discussed in the section of this lesson on potassium.

Recommended daily allowance...

The AI or Adequate Intake Level for adolescents and adults ages 14-50 years is 1.5 g sodium or 3.8 g salt per day for males and females, based on the amount needed to replace losses through sweat in moderately active people and to achieve a diet that provides sufficient amounts of other essential nutrients .



Foods high in sodium and salt...

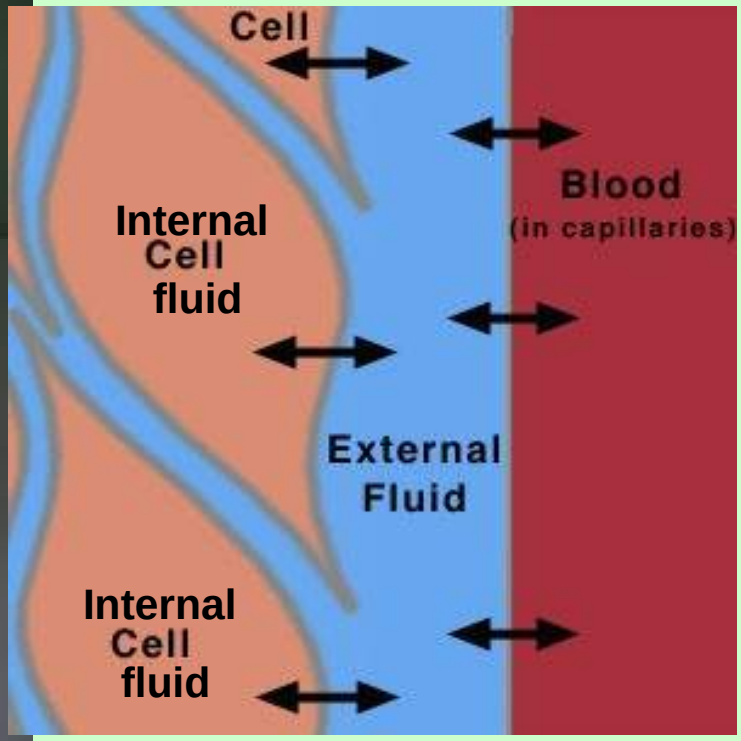


Food	Serving	Sodium (g)	Salt (g)
Hot dog, beef	**1**	**0.51**	**1.3**
Dill pickle	**1 spear**	**0.30**	**0.8**
Fish sandwich w/ tartar sauce	**1 sandwich**	**0.94**	**2.4**
Ham	**3 oz.**	**1.0**	**2.5**
Pretzels	**10 (2 oz.)**	**1.0**	**2.5**
Chicken Noodle Soup (canned)	**1 cup**	**1.4**	**3.4**
Macaroni & Cheese (canned)	**1 cup**	**1.3**	**3.3**
Potato Chips	**8 oz. bag**	**1.2**	**3.0**

Introducing potassium...

Like sodium, potassium is a cation : K+ This positively charged ion is in the fluid inside the cell, while sodium (NA+) was the principal cation in the fluid outside of the cell.





Sources of potassium...

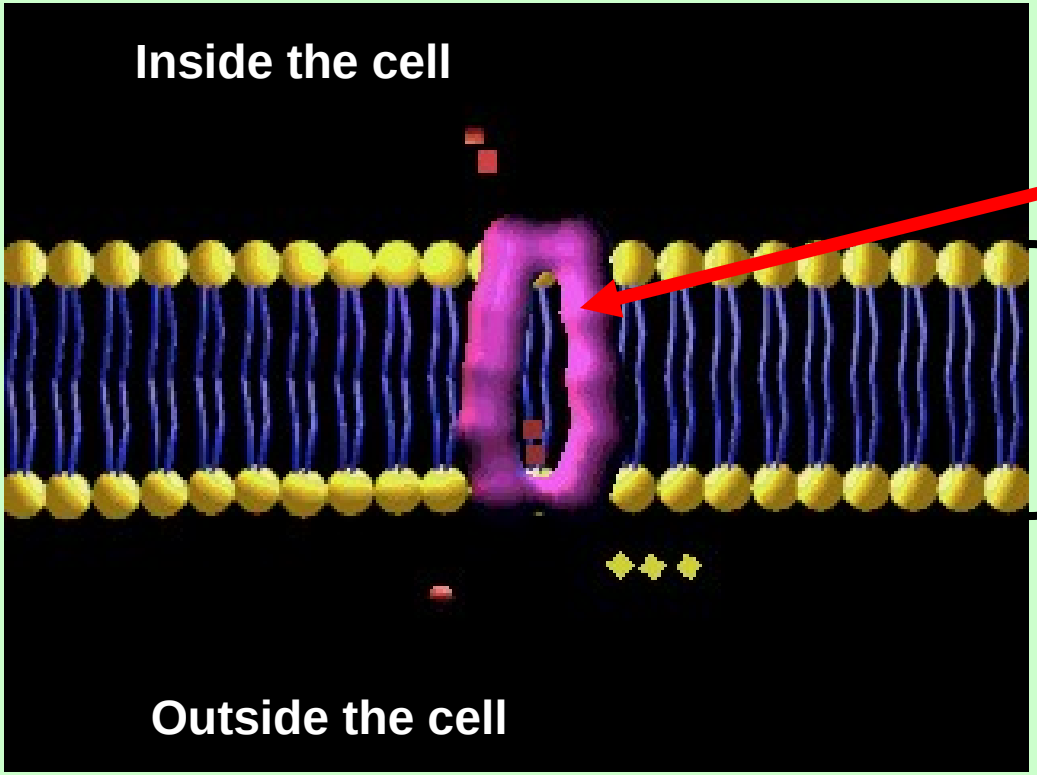
The richest sources of potassium are fruits and vegetables and nuts. Examples: 1 banana contains 422 mg., 1 baked potato with skin contains 926 mg., ½ cup of raisins contain 598 mg., 1 medium tomato contains 292 mg., 1 oz. almonds contain 200 mg.





Functions of potassium...

Potassium concentrations are about 30 times higher inside than outside cells, while sodium concentrations are more than ten times lower inside than outside cells. The concentration differences between potassium and sodium across the cell membranes create an electrochemical gradient or balance known as the 'membrane potential'. <u>Potassium helps maintain membrane potential</u>.



Enzyme pump or "gate"

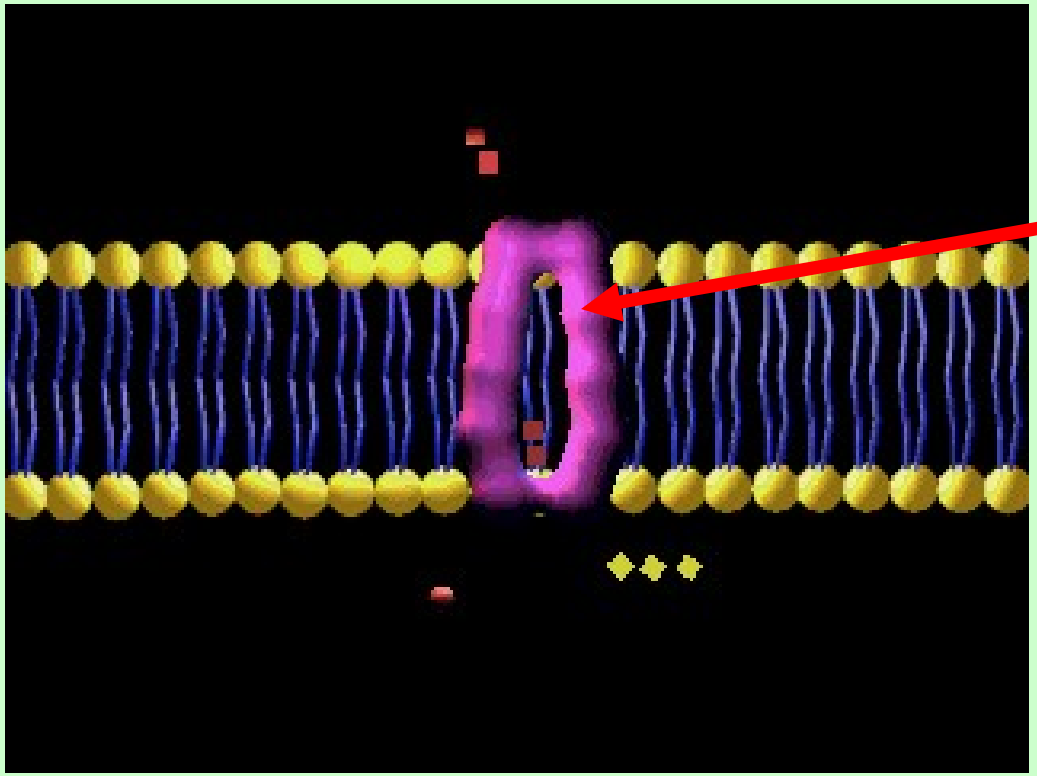
Semi-permeable cell membrane

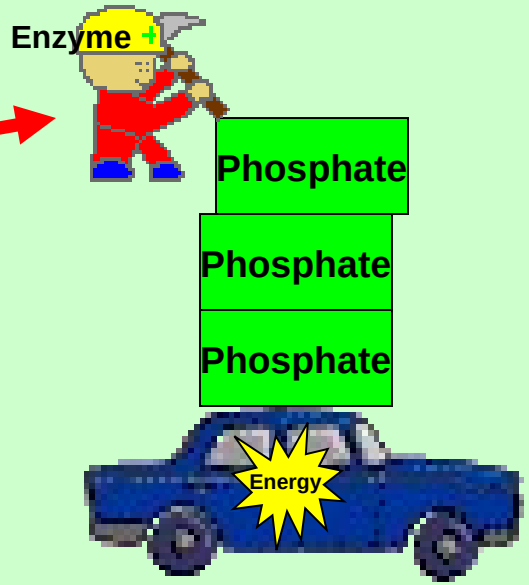
Potassium molecules

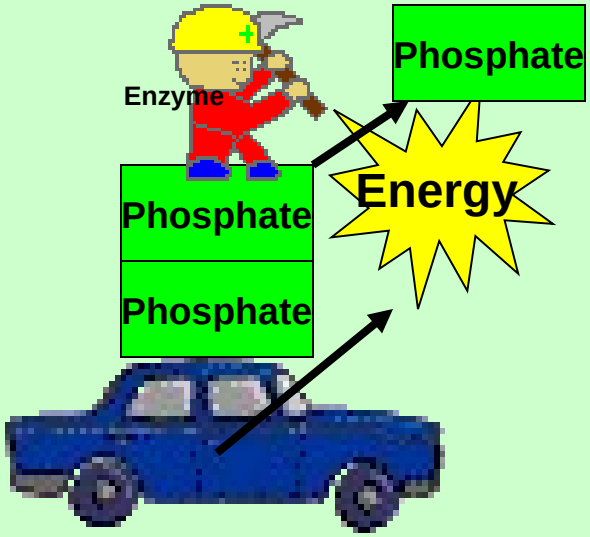
Sodium chloride molecules

Free-floating phosphate used to make ATP energy

A cell's 'membrane potential' or balance is maintained by ion pumps in the cell membrane, especially the sodium, potassium-ATPase pumps: Na+/K+ATPase . (the suffix "ase" indicates an enzyme and is pronounced 'ace') <u>Potassium and sodium serve as co-enzymes, with protein, to form an ion pump in the cell membrane.</u>

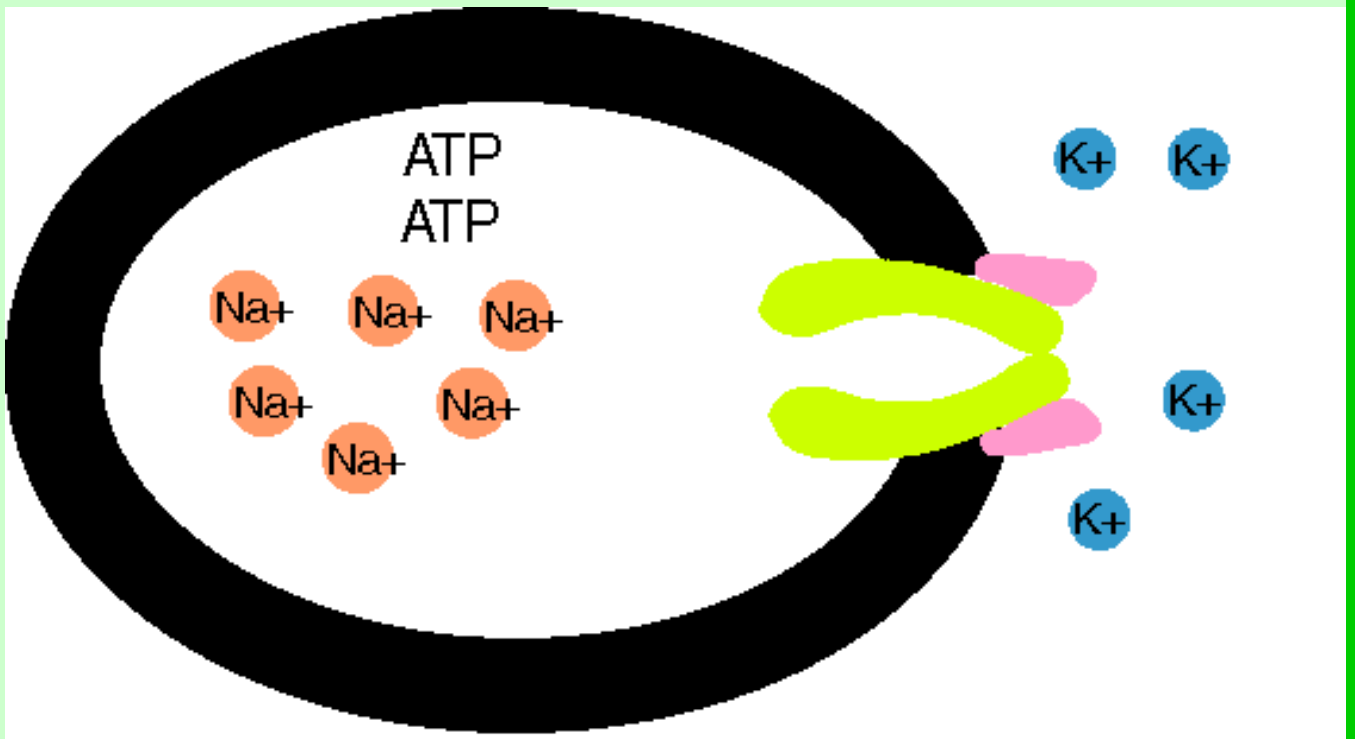






As sodium is pumped out of the cell in exchange for potassium, the enzyme pumps release or use ATP (energy). Their activity has been estimated to account for 20%-40% of the resting energy expenditure in a typical adult.

The large proportion of energy dedicated to maintaining sodium/potassium concentration gradients emphasizes the importance of this function in sustaining life.



When sodium chloride and potassium are dissolved in bodily fluids, they have a mild electrical charge. For this reason, they are called ELECTROLYTES. In the correct balance, these fluids are necessary for normal functioning of nerve transmissions and muscle contractions.







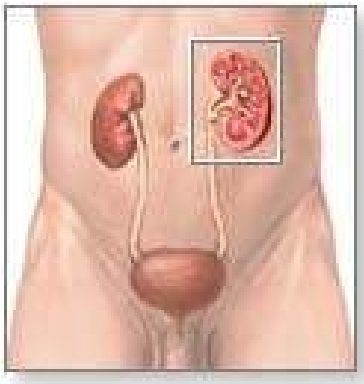
A beverage called an 'electrolyte solution' simply means that it contains a good balance of sodium chloride and potassium, as well as other electrolytes such as magnesium and calcium. Pedialyte is a brand name example, often given to children with diarrhea. Gatorade is a brand name example, often given to adults involved in heavy exercise who are sweating profusely.

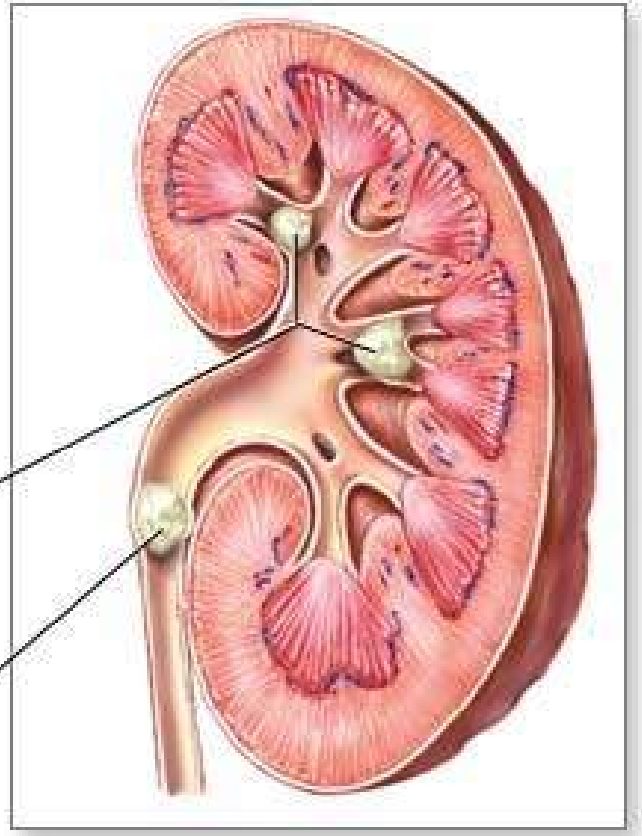
Recommended daily allowance...



The adequate intake level (AI) for potassium is based on intake levels that have been found to lower blood pressure, reduce salt sensitivity, and minimize the risk of kidney stones.

Adolescents and adults, both male and female, need 4.7 grams or 4,700 milligrams per day.





Potassium deficiencies...

An abnormally low plasma potassium concentration is referred to as hypokalemia, and is most commonly a result of prolonged vomiting, the use of some diuretics, overuse of laxatives, anorexia nervosa or bulimia, alcoholism, some forms of kidney disease and metabolic disturbances. Symptoms include fatigue, muscle weakness and cramps, and intestinal paralysis, which may lead to bloating, constipation, and abdominal pain. Severe hypokalemia may result in muscular paralysis or abnormal heart rhythms (cardiac arrhythmias) that can be fatal.





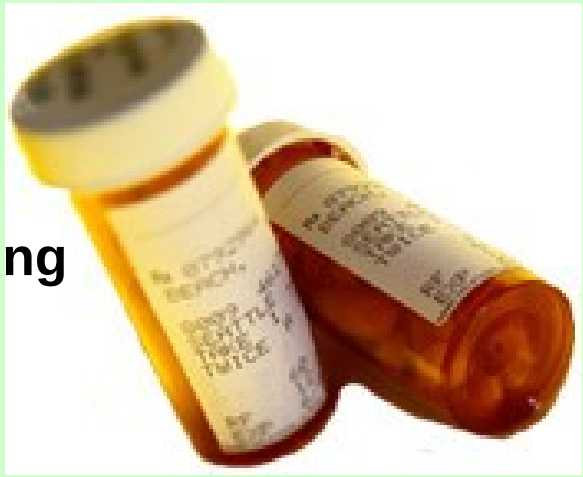


Black licorice candy contains a compound that increases urinary excretion of potassium. Use caution in habitually eating large amounts.

Potassium and drug interactions...



Potassium levels are often affected by over-the-counter and prescription drugs. Individuals with abnormal kidney function or on certain medications should be closely monitored. Common side effects of high blood pressure medications are increased frequency of urination and increased urinary excretion of potassiu





Common medications affecting potassium levels include:

Anti-coagulants
High-dose antibiotics
Decongestants
Cardiac and blood pressure medications
Diuretics NSAIDS (non-steroid anti-inflammatory drugs)

Introducing calcium...

Calcium (Ca) is the most common mineral in the human body. About 99% of the calcium in the body is found in bones and teeth, while the other 1% is found in the blood and soft tissue.





Calcium levels in the blood and fluid surrounding the cells (extracellular fluid) must be maintained within a very narrow concentration range for normal physiological functioning. <u>The physiological functions of calcium are so vital to survival that the body will de-mineralize bone to maintain normal blood calcium levels when calcium intake is inadequate.</u>

Sources of calcium…



Dairy products contain high levels of absorbable calcium, and provide nearly 75% of the calcium in American diets.

Calcium rich plants in the kale family (such as broccoli, bok choy, cabbage, mustard, and turnip greens) contain calcium that is as absorbable as that in milk, but also contain some food components that have been found to inhibit the absorption of calcium. Oxalic (ŏk-săl'ĭk) acid, also known as oxalate, is the most potent inhibitor of calcium absorption and is found at high concentrations in spinach and rhubarb and somewhat lower concentrations in sweet potatoes and dried beans.

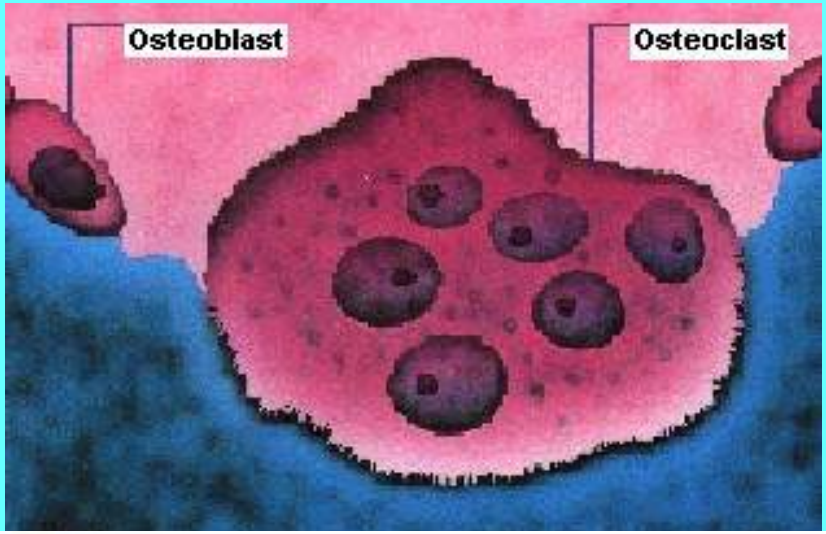
Recommended daily allowance...

Recommended Adequate Intake

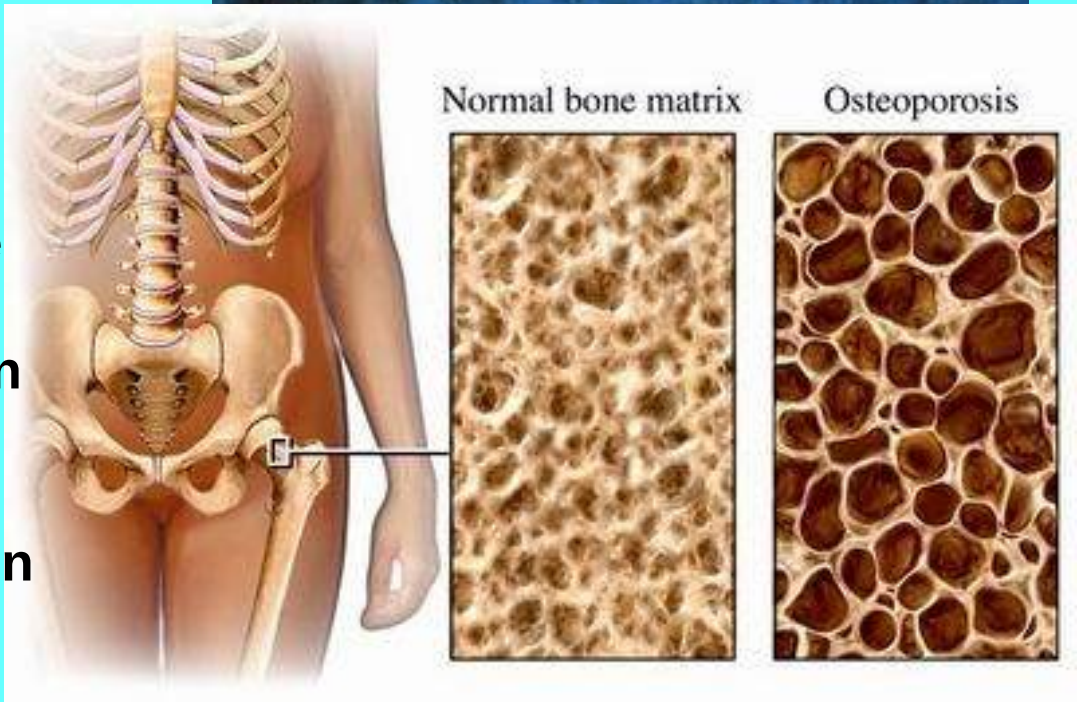
Male and Female Age	Calcium (mg/day)	Tolerable Upper Intake Level
0 to 6 months	210	
7 to 12 months	270	
1 to 3 years	500	2,500 mg/day
4 to 8 years	800	2,500 mg/day
14 to 18 years	1300	2,500 mg/day
19 to 50 years	1000	2,500 mg/day
51+ years	1200	

Average dietary intakes of calcium in the U.S. are well below the adequate intake (AI) recommendation for every age and gender group, especially in females. Only about 25% of boys and 10% of girls ages 9 to 17 are estimated to meet the AI recommendations.

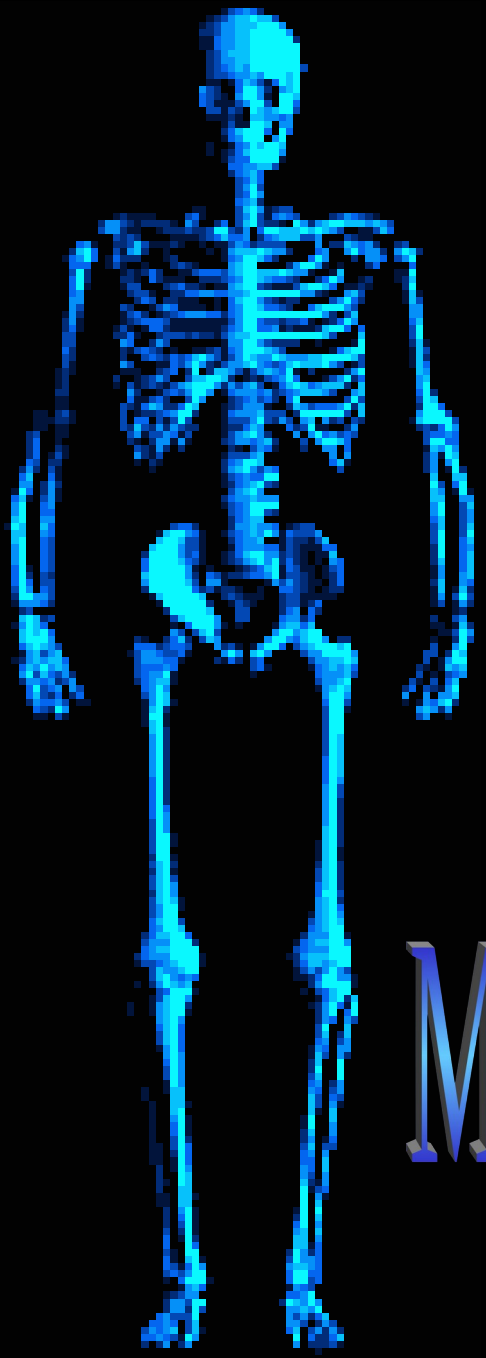
Functions of calcium...



Calcium is a major structural element in bones and teeth. Bone is a tissue that is remodeled throughout life. Bone cells called ‘osteoclasts’ begin the process of remodeling by dissolving or resorbing bone. Bone-forming cells called ‘osteoblasts’ then synthesize new bone to replace the bone that was resorbed. During normal growth, bone formation exceeds bone resorption. A condition known as ‘osteoporosis’ may result when bone resorption chronically exceeds formation.



Calcium also plays a significant role in the secretion of hormones such as insulin and as a cofactor with vitamin K for clotting blood.



THE END

MACROMINERALS

